

SoilFER ProDOC Alignment

Whether the analysis catalogue and build plan of Volume III can be delivered without breaching what FAO has promised USAID and Japan. Mostly yes — with one correction that matters, two obligations the plan did not know about, and four analyses missing from the catalogue.

DOCUMENTS	CHECKED AGAINST	DATE
GCP/GLO/1127/USA (SoilFER)	Volume III catalogue & build plan	2 September 2026
GCP/GLO/1242/JPN (SoilFER-VACS)		
RESULT		
Safe to proceed · 1 correction · 2 new obligations · 4 catalogue additions		

The plan is safe, and the catalogue is better aligned than I expected. Activity 1.6 of the SoilFER ProDOC names the exact analyses the project must deliver, and all twelve are already in the Volume III catalogue with the right units and methods. Activity 1.5 requires each country's National Reference Soil Laboratory to work within GLOSOLAN using "standardized methods and protocols" — which is precisely the GLOSOLAN-first default the catalogue ships. Nothing in either document names a data standard the LIMS itself must implement.

But one sentence changes the framing of Volume III's Phase 5. Section 2.4, Information Technology, commits the project's data products to being "fully integrated in ... Global Soil Information System (GloSIS)". I wrote that if GloSIS

never became worth doing, nothing would have been spent. That was wrong: it is a written commitment, and it will come due. The *architecture* I proposed still holds — arguably it holds better — but it must be presented as deferred to a scheduled point, not as optional.

Two obligations the build plan currently under-weights. Proficiency testing is a means of verification for Activity 1.1, and two FAO/IAEA ring trials are committed under Activity 1.6 — so the free-text QC blob flagged in Volume II is not merely a scientific weakness, it is a hole in a reportable indicator. And "at least 5 LIMS connected to NSIS" is a counted target under Activity 1.10, with the NSIS being built under a *separate* Letter of Agreement — meaning the exchange format is an interface contract to be negotiated, not a feature to be invented.

12/12 MANDATED ANALYSES Every Activity 1.6 analyte already in the catalogue	1 CORRECTION GloSIS is committed in writing, not optional	2 OBLIGATIONS Proficiency testing · NSIS interface as a counted target	4 ADDITIONS Gamma spectrometry, fertilizer nutrients, CRNS, legacy import
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01

The two projects, and where the LIMS sits

	SOILFER	SOILFER-VACS
Symbol	GCP/GL0/1127/USA	GCP/GL0/1242/JPN
Donor	USAID / US Government	Japan
Budget	USD 30 000 000	USD 6 000 000
Countries	Guatemala, Honduras, Zambia, Kenya, Ghana	Mozambique, Tunisia
Dates	17 May 2023 → 16 May 2027	1 April 2024 → 30 March 2025
Executing unit	FAO Land and Water Division (NSL), Rome	

The seven countries map exactly onto the seven laboratories hard-coded in the LIMS (koboController.js — GTM, HND, ZMB, GHA, KEN under S0ILFER-US; MOZ and TUN under S0ILFER-JPN). The codebase and the project documents agree, which is reassuring and worth saying.

One date needs checking. The VACS not-to-exceed date is 30 March 2025 — seventeen months before today. The LIMS still carries Mozambique and Tunisia as active laboratories under S0ILFER-JPN. Either the project was extended, or those two laboratories are now operating outside a live project envelope. That is a question for the project manager, not a finding, but the answer changes who pays for their support after go-live.

02

What the ProDOCs actually require of the LIMS

Six commitments reach the laboratory system. Two of them are counted indicators, which means someone will report a number against them.

COMMITMENT	SOURCE	REACHES THE LIMS AS	
"at least 5 Laboratory Information Management Systems (LIMS) connected to NSIS "	Activity 1.10 logframe target 2	A counted, reportable indicator. The SIS API is a deliverable.	COUNTED
Laboratories operating "with standardized SOPs and proficiency testing "; means of verification includes "proficiency testing results"	Activity 1.1 logframe target & MoV	PT/ILC rounds must be recordable and reportable.	COUNTED
"Analytical database and spectral libraries uploaded on Laboratory Information Management System (LIMS) "	Activity 1.6 means of verification	The LIMS is the evidence that the analysis happened.	EVIDENCE
LIMS features: "workflow	Activity 1.10 · VACS 1.4	No named standard.	OPEN

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and data tracking, quality assurance and control, and a standard architecture for data exchange"	narrative, both ProDOCs	Deliberately open — this is the clause that makes Volume III safe.	
"The data products will be fully integrated in ... the Global Soil Information System (GloSIS)"	Section 2.4 Information Technology	An eventual obligation, two steps downstream of the LIMS. See §03.	COMMITTED
"two ring trials will be carried out to follow up the analytical quality of the dry chemistry analyses" (with Joint FAO/IAEA Centre)	Activity 1.6 narrative	Inter-laboratory comparison rounds on spectral data.	COMMITTED
"5 national legacy data repositories"; collating, digitizing and harmonizing legacy data	Activity 1.2 logframe target	Import of historical results whose methods differ from today's.	GAP

03

The correction: GloSIS is committed, not optional

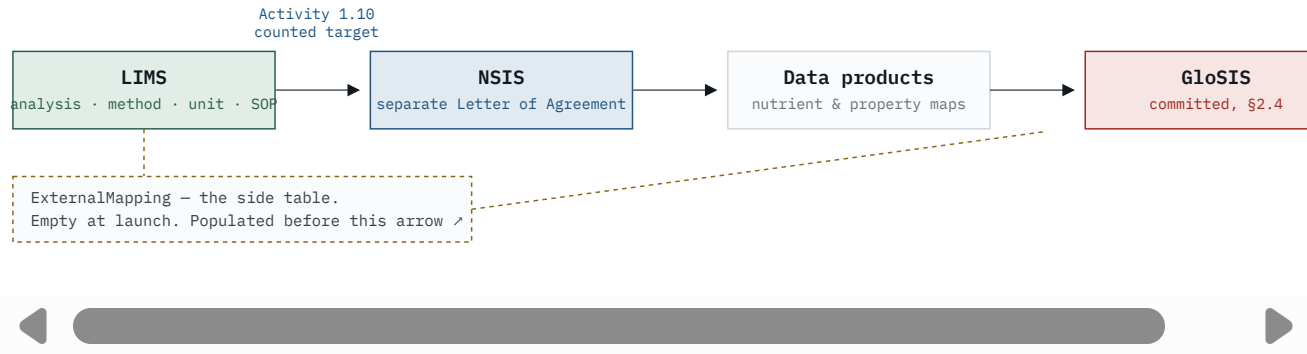
Volume III told you the ontology could be shelved and that if it never became worth doing, nothing would have been spent. The first half is right. The second half is not.

"The data products will be fully integrated in the FAO's corporate Geospatial Information system developed for the Hand-in-Hand Initiative; Global Soil Information System (GloSIS) developed by FAO's Global Soil Partnership."

SoilFER ProDOC, GCP/GL0/1127/USA – Section 2.4, Information Technology, p. 51

Read it precisely, because the precision is what saves the plan. The commitment attaches to **data products** — and the data products are defined in Activities 1.9 and 1.11 as the national and regional *maps*: soil nutrient maps, nutrient budget maps, soil property maps, thematic maps of threats and functions. It does not attach to the laboratory system. The LIMS's own obligation, in Activity 1.10, is the deliberately unnamed "standard architecture for data exchange" and a connection to the NSIS.

So the chain of obligation has three hops, and GloSIS sits at the far end of it:



GloSIS is two hops downstream of the laboratory. That is why deferring it is safe — and it is also exactly why a small mapping table, rather than ontology fields spread across the schema, is the right way to hold the obligation until it comes due.

COR-01 Correction to Volume III

Phase 5 is scheduled work, not speculative work

The design does not change. ExternalMapping as a side table with no UI remains the right structure, and moving the ten dormant glosis_* columns off Analysis and Methodology remains the right first step. What changes is the framing and the deadline: this is a commitment with a date attached, not an option to be exercised.

The practical consequence is small but real. The mapping table must be populated for the ~40 catalogue analyses that have a GloSIS property equivalent, and that population has to happen before national data products are handed to GloSIS — which, given a project end of 16 May 2027 and Activities 1.9 and 1.11 producing maps well before then, means the work should be scheduled into 2026, not left open-ended.

AMEND THE PLAN

Rewrite Phase 5's goal as: *"Park the ontology in a side table now; populate it against the catalogue before the first national data products go to GloSIS."* Add a dated milestone tied to the Activity 1.9 map deliverables. Delete the sentence "If it never becomes worth doing, nothing was spent" — it is not accurate against this ProDOC.

04 Two obligations the build plan under-weights

OBL-01 Counted indicator

Proficiency testing is reportable, and the LIMS cannot currently record it

Activity 1.1's target is a "Fully operational National Soil Laboratory Network in each beneficiary country, with **standardized SOPs and proficiency testing**", and its means of verification explicitly lists "proficiency testing results". Activity 1.6 additionally commits to "two ring trials ... to follow up the analytical quality of the dry chemistry analyses", run jointly with the Joint FAO/IAEA Centre.

Volume II found that `Batch.qcResults` is an unstructured JSON string with no certified values, no acceptance windows and no computed outcome. That was raised as a scientific weakness. Against these documents it is also a gap in a reportable indicator: there is nowhere in the system to hold a PT round, an assigned value, a laboratory's z-score, or a ring-trial result. Notably the repository already contains `tests/scenarios/pt_ilc.test.js`, so someone intended this once.

AMEND THE PLAN

Promote typed QC from Phase E to sit alongside Phase D, and add a distinct proficiency-testing model — round, provider, assigned value, uncertainty, the lab's result, z-score, and outcome — separate from internal batch QC. A PT round is not a QC batch; it is a laboratory-level competence record and it will be reported to FAO.

OBL-02 Counted indicator

The NSIS connection is a target, and the other side is a different contractor

Activity 1.10's second target is "at least 5 Laboratory Information Management Systems (LIMS) connected to NSIS". Someone will count them. Meanwhile the Letters of Agreement list "*LIMS Development (1): Laboratory Software and System*" and "*NSIS Development (1): SDI for Field Data Collection Tool, NSIS, NMS and DSS*" as **two separate agreements**.

That means the exchange format between them is an interface contract to be agreed with another party, not a design decision the LIMS team can make alone. The existing `/api/v1/sis` endpoints are a reasonable starting proposal — and Volume I's scoping fixes to them now matter more than they did — but they should be treated as a draft to be negotiated and versioned, not as the finished article.

There is a useful convergence here. The semantics that the NSIS will eventually need in order to push data products to GloSIS are the same semantics the `ExternalMapping` table is designed to hold. Filling that table is what lets the LIMS answer the NSIS in terms the NSIS can forward — which is the cheapest way to satisfy both COR-01 and this target at once.

AMEND THE PLAN

Add a Phase 6: publish a versioned LIMS ↔ NSIS data-exchange specification, agree it with the NSIS implementer, and validate it against one real country's data before it is called "connected". Budget for the fact that the other side may not exist yet.

05 Four things the catalogue is missing

ADD-01 Catalogue addition

Gamma-ray spectrometry is funded, trained and unrepresented

The ProDOC equips each National Reference Soil Laboratory with "soil spectroscopy **and gamma spectrometry**", funds Seibersdorf training courses on "MIRS and Gamma Spectrometry ... one week per topic, for 5 countries, total 15 technicians", and commits "9 follow-up expert missions of 1 week each in the field of MIRS and GRS to all five countries". Activity 1.6 adds that "further gamma spectrometry techniques can be introduced for in-situ soil property mapping (e.g., soil texture)".

Volume III's catalogue has MIR and vis-NIR and no GRS at all. Given that fifteen technicians are being trained on it and it is explicitly proposed for texture estimation — a property the LIMS already models — its absence would surface as soon as the first country starts using the instrument.

ADD

A gamma-ray spectrometry scan analysis alongside MIR and NIR, plus the derived radioelement concentrations (K-40, U-238 eq, Th-232 eq, total count) that GRS actually reports, and a provenance flag distinguishing GRS-estimated properties from measured ones.

ADD-02 Catalogue addition

These laboratories will analyse fertilizers, not only soil

Both ProDOCs commit to fertilizer quality analysis. SoilFER Activity 3.3 establishes "a capacity development programme on fertilizer quality analysis" with "particular attention to nutrient content *and* contaminant loading", and Activity 1.5's capacity programme explicitly includes "fertilizer quality analysis". The wider rationale — the fertilizer crisis, "the use of low-quality fertilizers that do not contain adequate proportion of nutrients and have high loads of contaminants" — is the founding argument of the project.

Volume III's catalogue covers the contaminant half (FERT_HM, heavy metals in phosphate fertilizer materials, GLOSOLAN 2024) and none of the nutrient half. A laboratory checking whether a bag of urea contains the nitrogen it claims has nowhere to put the result.

ADD

A fertilizer analysis category: total N, ammoniacal N, nitrate N, urea N, total and water-soluble P_2O_5 , neutral-ammonium-citrate-soluble P_2O_5 , water-soluble K_2O , S, Ca, Mg, micronutrients, moisture and particle size — with the sample type distinguishing fertilizer from soil, since the interpretation bands and plausible ranges are entirely different.

ADD-03 Interface question

Soil moisture from cosmic-ray neutron sensors needs a home

Activity 1.10's third and fourth targets are "at least 5 cosmic ray neutron sensors (CRNS) installed in 5 representative agro-ecological zones per country (in total at least 25)" and "at least 8 technicians trained per country on the use of CRNS". This is a continuous monitoring stream, not a laboratory determination.

It almost certainly belongs to the NSIS rather than the LIMS — but "almost certainly" is not a decision, and 25 sensors across seven countries will produce data that has to land somewhere with an owner. Worth settling explicitly rather than discovering later.

DECIDE

Confirm with the project manager that CRNS and WaPOR streams are NSIS responsibilities. If any part lands in the LIMS, it is a time-series concern and should not be forced into the sample-and-result model.

ADD-04 Catalogue addition

Legacy data import needs the method field the plan already built

Activity 1.2 targets "5 national legacy data repositories" and commits to "collating, digitizing, and **harmonizing** soil legacy data", which Activity 1.7 then incorporates into the national analytical databases. Harmonising historical soil data is almost entirely a question of knowing which method produced each number — a 1980s Walkley–Black carbon and a 2026 dry-combustion carbon are not the same measurement.

Volume III's method-bound result model is exactly what this needs, so the foundation is right. What is missing is the import path: a way to bring in historical results with a stated method, a stated unit, a stated basis and a provenance flag marking them as legacy rather than measured in this laboratory.

ADD

A legacy import with mandatory method and unit mapping per column, and provenance = IMPORTED on every row it creates — the same provenance field that SPC-01 needs for spectral predictions. One field, three uses.

06

Where the plan is already correct

Every mandated analysis is in the catalogue

Activity 1.6 names the analyses the project must perform. All twelve are present, with units and methods that match what the National Reference Laboratories will actually run:

PRODOC ACTIVITY 1.6 WORDING	CATALOGUE CODE	SHIPPED DEFAULT
percentage of coarse fragments	COARSE_FRAG	Gravimetric, sieved · ISO 11464
particle size distribution (clay, silt, sand)	SAND · SILT · CLAY	Pipette, dispersed · ISO 11277:2020
pH	PH	1:2.5 water · GLOSOLAN 2021
electrical conductivity	EC	1:5 water · GLOSOLAN 2021
SOC	SOC	Walkley–Black · GLOSOLAN 2021
carbonate content	CAC03	Volumetric calcimeter · GLOSOLAN 2021
phosphorous content	AVAIL_P	Olsen · GLOSOLAN 2021
total nitrogen content	TOTAL_N	Dumas · GLOSOLAN 2021
extractable potassium content	AVAIL_K	1 M NH ₄ OAc pH 7 · GLOSOLAN 2023

bulk density	BULK_DENS	Cylinder · GLOSOLAN 2024
cation exchange capacity	CEC	1 M NH ₄ OAc pH 7 · GLOSOLAN 2023
multispectral properties (dry chemistry)	SPEC_MIR · SPEC_NIR	Diffuse reflectance
"selected micronutrients and biodiversity indicators ... defined during the inception phase"	§G · §H	DTPA suite, hot-water B; MBC, respiration, enzymes

- **GLOSOLAN-first defaults are not a preference, they are the project design.** Activity 1.5 places each country's National Reference Soil Laboratory in GLOSOLAN and requires samples to be "processed and analyzed using standardized methods and protocols". Volume III's decision to default to the GLOSOLAN SOP and fall back to ISO is exactly this.
- **Spectral calibration services are committed, and the plan anticipated them.** Activity 1.7's target is "5 National Spectral Libraries and Calibration Services", using PLSR or SVM to predict N, P, K and SOC. Volume II's SPC-01 — add a provenance field before any prediction lands — is therefore not a hypothetical precaution. It is due before Activity 1.7 completes.
- **No named data standard binds the LIMS.** Both ProDOCs say "standard architecture for data exchange" and "standardizing formats and protocols for access and interoperability" without naming one. This is what makes the rollback defensible: the project asked for interoperability, not for a specific ontology, and the LIMS satisfies that through the NSIS interface.
- **The nine packages match how the project tiers its work.** Sampling is prioritised at 85 percent croplands, 10 percent grasslands, 5 percent forestlands, with a first campaign establishing an NPK baseline. Basic Fertility and Carbon Stock are precisely the panels that campaign needs.
- **Human-readable was the right instinct for this audience.** The project's own capacity-development activities train laboratory technicians and heads on "soil analysis (wet and dry chemistry), laboratory safety, laboratory data management, quality assurance and quality control" — not on linked data. A system whose configuration screens speak that vocabulary is aligned with the training the project is paying for.

Volume III's five phases survive. Three are amended and one is added.

PHASE	CHANGE	WHY
1 Seed the catalogue	Add fertilizer analyses and gamma spectrometry. Add sample type (soil / fertilizer / plant).	ADD-01, ADD-02
2 Defaults & wizard	Unchanged.	—
3 Method through to output	Add the provenance field here rather than in Phase E — measured / predicted / derived / imported.	ADD-04, SPC-01, Activity 1.7
4 Packages & derived values	Add typed QC <i>and</i> a separate proficiency-testing model. Add the legacy import path.	OBL-01, ADD-04
5 Ontology on the shelf	Reframed as scheduled work with a date, not an option. Mapping populated before data products reach GloSIS.	COR-01
6 NSIS interface NEW	Publish a versioned LIMS ↔ NSIS exchange specification; agree it with the NSIS implementer; validate on one country before declaring "connected".	OBL-02

On timing: SoilFER runs to 16 May 2027, roughly twenty months out. Volume III's six weeks plus these amendments is comfortably inside that, but the sequencing matters — the analytical database is the means of verification for Activity 1.6, so it has to be populated and credible well before the final evaluation, not at the end of it.